

Tax Multipliers in a DSGE Model*

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Abstract

This paper analyzes the stabilization properties of tax cuts in a DSGE model. We do this by simulating the a general equilibrium model with a rich set of tax levers in order to quantify tax multipliers over the short and medium-term. The novelty in this paper is that we focus on tax cuts that directly target consumption and investment, which are the most effective counter-cyclical fiscal policies. We find that the short-term multiplier associated with a cut to the consumption tax is larger than the multiplier associated with a cut to the labor income tax, and that the short-term multipliers associated with cuts to investment taxes are even larger. We compare these to the multipliers associated with cuts to broad taxes on capital and labor income, which are the focus of much of the existing literature. We also find that the medium-term multipliers associated with cuts to consumption and investment taxes are larger than the medium-term multipliers associated with cuts to broad taxes on capital and labor income. These results suggest that tax cuts that directly target consumption and investment are more effective counter-cyclical fiscal policies than tax cuts that target broad taxes on capital and labor income.

We study tax multipliers associated with taxes that directly stimulate private consumption spending or investment such as consumption taxes and investment tax credits. Consumption taxes have long been favored by tax economists for their relative efficiency vis-a-vis taxes on capital and labor. This paper attempts to argue that consumption taxes should also be favored by macroeconomists for their stabilization properties. Cuts to consumption taxes can quickly and effectively stimulate aggregate demand. The following analysis studies the quantitative impact of tax cuts in a dynamic stochastic general equilibrium (DSGE) model. In addition to consumption taxes, the model nests other tax policies, such as investment tax credits and changes to expensing and depreciation policies. These policies target investment by lowering the after-tax cost of capital. Historically, such instruments have been used by the U.S. to provide counter-cyclical fiscal policy (e.g. the Economic Recovery Act of 1981 increased the ability of businesses to take advantage of the investment tax credit). Despite their use, such policies have not been studied in a DSGE environment.

The goal of this paper is to study specific tax policies in a DSGE environment. A number of researchers have used DSGE models to study the government spending multiplier (e.g., Christiano, Eichenbaum and Rebelo (2010)), but less work has been done on tax multipliers. Those who provide insights on tax multipliers (e.g., Zubairy (2010), Kumhof, Coenen, Muir, Freedman, Mursula, Erceg, Furceri, Lalonde, Linde, Mourougane, Roberts,

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Laxton, de Resende, Roeger, Snudden, Trabandt and in 't Veld (2010)) focus on multipliers associated with broad taxes on capital and labor income. Such tax cuts are generally not the most effective counter-cyclical policies. We are the first to provide quantitative insights into the size of the multipliers associated with tax cuts that directly target consumption and investment.

Some related work to mention:

1. Kumhof et al. (2010) shows impulse response functions to a consumption tax cut. This paper seems to find that consumption taxes do well compared to other taxes and transfers, but don't do as well as gov't consumption and gov't investment
2. Leeper, Plante and Traum (2010) have consumption taxes, but don't allow them to vary with gov't debt
3. Zubairy (2010) doesn't not have consumption taxes, only capital and labor income taxes. Says few even have distortionary taxes in models of multipliers.
4. Traum and Yang (2010) allow for consumption taxes and allow them to vary. Not answering the same question as I want, but good to look at what they say about consumption taxes. In particular, how with cons taxes the consumer and producer price index will vary and what crowding out happens with consumption tax changes.
5. Coenen and Straub (2005) have consumption taxes, but don't use them as a policy instrument. They don't say anything interesting about consumption taxes.
6. Christiano et al. (2010) don't have consumption taxes and only focus on spending multipliers. I don't even think they have taxes. Fiscal and monetary policy rules look simple though- might want to take these.

The remainder of the paper is organized as follows. Section 1 describes the model. Section 1.7 describes the calibration and estimation strategy we use to identify parameter values. Section ?? presents the results. Section ?? concludes.

1 Model

1.1 Households

The economy is populated by a continuum of identical households. Each household maximizes expected lifetime utility over its habit-adjusted private consumption surplus $x_{c,t}$, hours worked h_t , and habit-adjusted government consumption surplus $x_{g,t}$:

$$E_0 \sum_{t=0}^{\infty} \beta^t d_t \left\{ \frac{[x_{c,t}^a (1 - h_t)^{1-a}]^{1-\gamma}}{1 - \gamma} + V(x_{g,t}) \right\} \quad (1.1)$$

where $\beta \in (0, 1)$ is the discount factor, d_t is an intertemporal preference shock, γ is the coefficient of relative risk aversion, and $a \in (0, 1)$ is the consumption expenditure share calibrated at the steady state. Utility over government consumption $V(x_{g,t})$ is separable from private utility, so government spending does not directly affect the marginal utility of private consumption or leisure. Following Zubairy (2010), the model incorporates deep habit formation in both private and government consumption as in ?, which generates

countercyclical markups and allows private consumption to rise in response to a government spending shock.

The habit-adjusted consumption surpluses are:

$$x_{c,t} = c_t - b_c s_{C,t-1} \quad (1.2)$$

$$x_{g,t} = g_t - b_g s_{G,t-1} \quad (1.3)$$

where $b_c \in [0, 1)$ and $b_g \in [0, 1)$ govern the degree of deep habit formation in private and government consumption. The habit stocks evolve according to:

$$s_{C,t} = \theta^c s_{C,t-1} + (1 - \theta^c) c_t \quad (1.4)$$

$$s_{G,t} = \theta^g s_{G,t-1} + (1 - \theta^g) g_t \quad (1.5)$$

where $\theta^c, \theta^g \in [0, 1)$ govern the speed of habit stock adjustment.

Households supply labor, accumulate physical capital, hold government bonds, and receive dividend income from firms. The period-by-period real budget constraint is:

$$\begin{aligned} (1 + \tau_t^c) c_t + b_t + i_t = & \frac{[1 + (R_{t-1} - 1)(1 - \tau_t^i)] b_{t-1}}{\pi_t} + (1 - \tau_t^w) w_t h_t \\ & + (1 - \tau_t^k) r_t^k u_t k_{t-1} + \tau_t^k \delta^\tau k_{t-1}^\tau + \tau_t^k e_t^\tau i_t + \tau_t^{ic} i_t + (1 - \tau_t^d) \text{div}_t + x_t \end{aligned} \quad (1.6)$$

where b_t denotes real bond holdings, R_t is the gross nominal interest rate, $\pi_t \equiv P_t/P_{t-1}$ is gross price inflation, w_t is the real wage, r_t^k is the real rental rate of capital, u_t is the capital utilization rate, div_t are real firm dividends, and x_t is a lump-sum government transfer. The tax instruments are: τ_t^c (consumption tax), τ_t^w (labor income tax), τ_t^k (capital income tax), τ_t^i (interest income tax), τ_t^d (dividend income tax), and τ_t^{ic} (investment tax credit). Only the interest component $(R_{t-1} - 1)$ of the gross bond return is subject to the interest income tax. Capital income is measured net of tax depreciation deductions $\tau_t^k \delta^\tau k_{t-1}^\tau$, where δ^τ is the tax depreciation rate and k_t^τ is the tax basis of the capital stock, and net of immediate expensing $\tau_t^k e_t^\tau i_t$, where e_t^τ is the expensing rate.

Physical capital accumulates according to:

$$k_t = (1 - \delta) k_{t-1} + i_t \left[1 - S \left(\frac{\mu_t i_t}{i_{t-1}} \right) \right] \quad (1.7)$$

where δ is the constant economic depreciation rate and μ_t is a stochastic investment efficiency shock. Following Smets and Wouters (2003) and Zubairy (2010), the investment adjustment cost function takes the form $S \left(\frac{\mu_t i_t}{i_{t-1}} \right) = \frac{\kappa}{2} \left(\frac{\mu_t i_t}{i_{t-1}} - 1 \right)^2$, so that in the deterministic steady state $S = S' = 0$ and $S'' = \kappa > 0$.

The tax basis of the capital stock evolves as:

$$k_t^\tau = (1 - \delta^\tau) k_{t-1}^\tau + i_t (1 - e_t^\tau) \quad (1.8)$$

Only the non-expensed fraction $(1 - e_t^\tau)$ of new investment is added to the tax basis; the immediately expensed portion is deducted in the period of investment.

Household optimality conditions

The first-order condition with respect to consumption defines the marginal utility of income λ_t :

$$\lambda_t = \frac{d_t \cdot a \cdot [x_{c,t}^a (1 - h_t)^{1-a}]^{1-\gamma}}{x_{c,t} (1 + \tau_t^c)} \quad (1.9)$$

The consumption tax τ_t^c drives a wedge between the marginal utility of income and the marginal utility of the consumption surplus: a higher τ_t^c lowers λ_t for any given level of $x_{c,t}$ and h_t .

The intratemporal labor-leisure condition equates the marginal disutility of work to the after-markup real wage:

$$d_t (1 - a) \frac{[x_{c,t}^a (1 - h_t)^{1-a}]^{1-\gamma}}{1 - h_t} = \frac{\lambda_t w_t}{\tilde{w}_t} \quad (1.10)$$

where $\tilde{w}_t$ is the wage markup arising from monopolistic competition in the labor market.

The optimal utilization rate equates the after-tax marginal product of capital to the marginal cost of utilization:

$$(1 - \tau_t^k) r_t^k = \gamma_1 u_t^{\sigma_u} \quad (1.11)$$

where $\gamma_1 = (1 - \tau_k^{ss}) r_k^{ss}$ is pinned down at the steady state and σ_u governs the convexity of the utilization cost function $a(u_t) = \frac{\gamma_1}{1 + \sigma_u} (u_t^{1 + \sigma_u} - 1)$. Relative to Zubairy (2010), the term $\delta q_t \tau_t^k$ is absent from this condition because depreciation deductions are now tracked through the shadow value of the tax basis $v_{\tau,t}$ defined below, avoiding double-counting.

The capital Euler equation governs the optimal capital stock:

$$\lambda_t q_t = \beta E_t \lambda_{t+1} \left[(1 - \tau_{t+1}^k) r_{t+1}^k u_{t+1} - \frac{\gamma_1}{1 + \sigma_u} (u_{t+1}^{1 + \sigma_u} - 1) + q_{t+1} (1 - \delta) \right] \quad (1.12)$$

where q_t is Tobin's marginal q . Similarly, the term $\delta q_{t+1} \tau_{t+1}^k u_{t+1}$ present in Zubairy (2010) is omitted here for the same reason.

The investment Euler equation governs the optimal investment decision:

$$\lambda_t (1 - \tau_t^{ic} - \tau_t^k e_t^\tau) = \lambda_t q_t \left[1 - S\left(\frac{\mu_t i_t}{i_{t-1}}\right) - \frac{\mu_t i_t}{i_{t-1}} S'\left(\frac{\mu_t i_t}{i_{t-1}}\right) \right] + \beta E_t \lambda_{t+1} q_{t+1} \left(\frac{\mu_{t+1} i_{t+1}}{i_t} \right)^2 S'\left(\frac{\mu_{t+1} i_{t+1}}{i_t}\right) + \lambda_t v_{\tau,t} (1 - e_t^\tau) \quad (1.13)$$

The left-hand side is the net cost of investing one unit: the investment tax credit τ_t^{ic} and the immediate expensing saving $\tau_t^k e_t^\tau$ each reduce the effective outlay. The term $\lambda_t v_{\tau,t} (1 - e_t^\tau)$ on the right-hand side is the present value of future tax depreciation deductions on the non-expensed share of investment. The shadow value of the tax basis $v_{\tau,t}$ satisfies its own Euler equation:

$$\lambda_t v_{\tau,t} = \beta E_t \lambda_{t+1} \left[(1 - \delta^\tau) v_{\tau,t+1} + \tau_{t+1}^k \delta^\tau \right] \quad (1.14)$$

An additional unit of tax basis yields a deduction $\tau_{t+1}^k \delta^\tau$ next period and a depreciated stock $(1 - \delta^\tau) v_{\tau,t+1}$ of future deductions.

Finally, the bond Euler equation is:

$$\lambda_t = \beta E_t [1 + (R_t - 1)(1 - \tau_{t+1}^i)] \frac{\lambda_{t+1}}{\pi_{t+1}} \quad (1.15)$$

Only the interest earned $(R_t - 1)$ on bond holdings is subject to the interest income tax τ_{t+1}^i , so the relevant return is the after-tax net interest. This differs from Zubairy (2010), in which the pre-tax gross return R_t governs the bond Euler.

1.2 Firms

1.2.1 Final Goods Producers

A continuum of monopolistically competitive firms indexed by $i \in [0, 1]$ each produce a differentiated variety of output demanded by households, the government, and for investment. Because households and the government form deep habits over individual varieties (Section 1), the demand for variety i in the private consumption and government markets is:

$$c_{i,t} = \left(\frac{P_{i,t}}{P_t} \right)^{-\eta} x_{c,t} + b_{cSC,i,t-1} \quad (1.16)$$

$$g_{i,t} = \left(\frac{P_{i,t}}{P_t} \right)^{-\eta} x_{g,t} + b_{gSG,i,t-1} \quad (1.17)$$

Investment demand for variety i is $i_{i,t} = (P_{i,t}/P_t)^{-\eta}(y_t - c_t - g_t)$; there are no deep habits in investment. Here $\eta > 1$ is the elasticity of substitution across varieties and P_t is the aggregate price index. The habit-inelastic components $b_{cSC,i,t-1}$ and $b_{gSG,i,t-1}$ make a portion of demand perfectly price-inelastic, generating countercyclical markups: when aggregate demand expands, the price-elastic share rises, increasing effective demand elasticity and compressing markups. Firms therefore invest in their customer base by holding prices low today to earn higher profits on a larger habit stock in the future.

1.2.2 Intermediate Goods Producers

Each firm i produces output using capital services $u_t k_{i,t-1}$ and labor $h_{i,t}$ according to a Cobb-Douglas technology with a fixed operating cost ψ :

$$\tilde{y}_{i,t} = z_t (u_t k_{i,t-1})^\theta h_{i,t}^{1-\theta} - \psi \quad (1.18)$$

where z_t is an aggregate technology shock, θ is the capital income share, and ψ is calibrated to match a realistic profit-to-output ratio in steady state. The technology shock follows an AR(1) process:

$$\log z_t = \rho_z \log z_{t-1} + \varepsilon_t^z, \quad \varepsilon_t^z \sim \mathcal{N}(0, \sigma_z^2) \quad (1.19)$$

where $\rho_z \in [0, 1)$. Cost minimization at given factor prices yields the symmetric-equilibrium factor demand conditions:

$$w_t = mc_t \cdot z_t (1 - \theta) (u_t k_{t-1})^\theta h_t^{-\theta} \quad (1.20)$$

$$r_t^k = mc_t \cdot z_t \theta (u_t k_{t-1})^{\theta-1} h_t^{1-\theta} \quad (1.21)$$

where mc_t is the common real marginal cost across firms.

Following Rotemberg (1982), each firm faces a quadratic cost of changing its nominal price, which reduces output available for final demand. In symmetric equilibrium, aggregate output net of price adjustment costs is:

$$y_t = z_t(u_t k_{t-1})^\theta h_t^{1-\theta} - \psi - \frac{\alpha_P}{2}(\pi_t - \bar{\pi})^2 \quad (1.22)$$

where $\alpha_P > 0$ governs the magnitude of price adjustment costs and $\bar{\pi}$ is the steady-state gross inflation rate.

Because deep habits create dynamic pricing incentives, each firm tracks the shadow value of its customer base in each demand segment. In symmetric equilibrium, the shadow values $\tilde{v}_t^c$, $\tilde{v}_t^g$, and $\tilde{v}_t^i$ for the private consumption, government consumption, and investment markets satisfy:

$$1 - mc_t - \tilde{v}_t^i = 0 \quad (1.23)$$

$$\frac{1 - mc_t - \tilde{v}_t^c}{\theta^c - 1} = \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \left[b_c \tilde{v}_{t+1}^c + \frac{\theta^c}{\theta^c - 1} (1 - mc_{t+1} - \tilde{v}_{t+1}^c) \right] \quad (1.24)$$

$$\frac{1 - mc_t - \tilde{v}_t^g}{\theta^g - 1} = \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \left[b_g \tilde{v}_{t+1}^g + \frac{\theta^g}{\theta^g - 1} (1 - mc_{t+1} - \tilde{v}_{t+1}^g) \right] \quad (1.25)$$

Equation (1.23) states that the investment market, lacking habit-inelastic demand, yields a zero effective markup. Equations (1.24)–(1.25) are forward-looking Euler equations: a firm accepts a lower current markup to build its habit stock, earning higher future rents.

These conditions together with the demand structure yield the aggregate price Phillips curve:

$$\eta(\tilde{v}_t^c x_{c,t} + \tilde{v}_t^g x_{g,t} + \tilde{v}_t^i (y_t - c_t - g_t)) + \alpha_P \pi_t (\pi_t - \bar{\pi}) - y_t = \alpha_P \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \pi_{t+1} (\pi_{t+1} - \bar{\pi}) \quad (1.26)$$

Each household is also a monopolistic supplier of a differentiated labor variety and faces quadratic wage adjustment costs. Defining gross wage inflation $\pi_t^w \equiv (w_t/w_{t-1})\pi_t$, the wage Phillips curve is:

$$\pi_t^w = \frac{w_t}{w_{t-1}} \pi_t \quad (1.27)$$

$$(\eta_w - 1)(1 - \tau_t^w)h_t + \alpha_W \pi_t^w (\pi_t^w - \bar{\pi}) - \frac{\eta_w h_t}{\tilde{w}_t} = \alpha_W \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \pi_{t+1}^w (\pi_{t+1}^w - \bar{\pi}) \quad (1.28)$$

where $\eta_w > 1$ is the elasticity of substitution across labor varieties and α_W governs wage adjustment costs. The labor income tax τ_t^w enters the wage Phillips curve because it affects the effective value of the marginal product of labor to the household.

Firm dividends are the residual of net output after paying all factors:

$$\text{div}_t = y_t - w_t h_t - r_t^k u_t k_{t-1} \quad (1.29)$$

The aggregate resource constraint equates gross production (net of fixed costs) to all expenditure uses plus utilization and price and wage adjustment costs:

$$z_t(u_t k_{t-1})^\theta h_t^{1-\theta} - \psi = c_t + g_t + i_t + \frac{\gamma_1}{1 + \sigma_u} (u_t^{1+\sigma_u} - 1) k_{t-1} + \frac{\alpha_P}{2} (\pi_t - \bar{\pi})^2 + \frac{\alpha_W}{2} (\pi_t^w - \bar{\pi})^2 w_t \quad (1.30)$$

1.3 Government/Fiscal Authority

The government issues one-period nominal bonds, purchases goods, pays lump-sum transfers to households, and collects tax revenue from all six tax instruments. The real government budget constraint is:

$$b_t = \frac{R_{t-1} b_{t-1}}{\pi_t} + g_t + x_t - taxrev_t \quad (1.31)$$

where b_t is real government debt, g_t is real government purchases, x_t is real lump-sum transfers, and $taxrev_t$ is total real tax revenue. Tax revenue is:

$$\begin{aligned} taxrev_t = & \tau_t^c c_t + \tau_t^w w_t h_t + \tau_t^k \left(r_t^k u_t k_{t-1} - \delta^\tau k_{t-1}^\tau - e_t^\tau i_t \right) \\ & + \tau_t^i (R_{t-1} - 1) \frac{b_{t-1}}{\pi_t} + \tau_t^d \text{div}_t - \tau_t^{ic} i_t \end{aligned} \quad (1.32)$$

The six components are: consumption tax revenue ($\tau_t^c c_t$), labor income tax revenue ($\tau_t^w w_t h_t$), capital income tax revenue net of tax depreciation deductions and immediate expensing ($\tau_t^k(\cdot)$), interest income tax revenue on bond interest ($\tau_t^i (R_{t-1} - 1) b_{t-1} / \pi_t$), dividend tax revenue ($\tau_t^d \text{div}_t$), and investment tax credit outlays ($-\tau_t^{ic} i_t$).

Fiscal rules

Each fiscal instrument follows an autoregressive rule that allows feedback from lagged government debt and lagged output, following Zubairy (2010). Debt feedback ensures intertemporal fiscal solvency; output feedback captures automatic stabilizers. Because g_t and x_t are expenditures that the government cuts when debt is elevated, their debt-feedback coefficients enter with a negative sign. Tax rates carry a positive debt-feedback sign, rising to close the fiscal gap.

Government purchases and lump-sum transfers follow:

$$\log\left(\frac{g_t}{\bar{g}}\right) = \rho_g \log\left(\frac{g_{t-1}}{\bar{g}}\right) - \rho_{g,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{g,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^g \quad (1.33)$$

$$\log\left(\frac{x_t}{\bar{x}}\right) = \rho_x \log\left(\frac{x_{t-1}}{\bar{x}}\right) - \rho_{x,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{x,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^x \quad (1.34)$$

The rules for the capital and labor income tax rates, which follow Zubairy (2010), are:

$$\log\left(\frac{\tau_t^k}{\bar{\tau}^k}\right) = \rho_k \log\left(\frac{\tau_{t-1}^k}{\bar{\tau}^k}\right) + \rho_{k,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{k,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^k \quad (1.35)$$

$$\log\left(\frac{\tau_t^w}{\bar{\tau}^w}\right) = \rho_w \log\left(\frac{\tau_{t-1}^w}{\bar{\tau}^w}\right) + \rho_{w,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{w,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \phi \varepsilon_t^k + \varepsilon_t^w \quad (1.36)$$

The term $\phi \varepsilon_t^k$ in the labor tax rule allows capital tax shocks to generate correlated movements in the labor tax rate, capturing the empirical co-movement between capital and labor income tax changes.

The four extension fiscal rules for the additional tax instruments take the same log-ratio form:

$$\log\left(\frac{\tau_t^c}{\bar{\tau}^c}\right) = \rho_c \log\left(\frac{\tau_{t-1}^c}{\bar{\tau}^c}\right) + \rho_{c,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{c,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^c \quad (1.37)$$

$$\log\left(\frac{\tau_t^i}{\bar{\tau}^i}\right) = \rho_i \log\left(\frac{\tau_{t-1}^i}{\bar{\tau}^i}\right) + \rho_{i,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{i,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^i \quad (1.38)$$

$$\log\left(\frac{\tau_t^d}{\bar{\tau}^d}\right) = \rho_d \log\left(\frac{\tau_{t-1}^d}{\bar{\tau}^d}\right) + \rho_{d,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{d,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \phi_{wd} \varepsilon_t^w + \varepsilon_t^d \quad (1.39)$$

The dividend tax rule includes $\phi_{wd} \varepsilon_t^w$, capturing the empirical co-movement between changes in labor and dividend income tax rates. Because the steady-state values of the investment tax credit $\bar{\tau}^{ic}$ and the expensing rate $\bar{e}^\tau$ are calibrated to zero, their rules are specified in level-deviation form to avoid undefined log-ratios:

$$\tau_t^{ic} - \bar{\tau}^{ic} = \rho_{ic}(\tau_{t-1}^{ic} - \bar{\tau}^{ic}) + \rho_{ic,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{ic,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^{ic} \quad (1.40)$$

$$e_t^\tau - \bar{e}^\tau = \rho_e(e_{t-1}^\tau - \bar{e}^\tau) + \rho_{e,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{e,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^e \quad (1.41)$$

1.4 Monetary Authority

The monetary authority sets the gross nominal interest rate R_t according to a Taylor rule with interest rate smoothing, following Zubairy (2010):

$$\log\left(\frac{R_t}{\bar{R}}\right) = \alpha_R \log\left(\frac{R_{t-1}}{\bar{R}}\right) + (1 - \alpha_R) \left[\alpha_\pi \log\left(\frac{\pi_t}{\bar{\pi}}\right) + \alpha_Y \log\left(\frac{y_t}{\bar{y}}\right) \right] + \varepsilon_t^m \quad (1.42)$$

where $\alpha_R \in (0, 1)$ governs interest rate inertia, $\alpha_\pi > 1$ is the response to inflation deviations from steady state (satisfying the Taylor principle), $\alpha_Y > 0$ is the response to output deviations, and ε_t^m is a monetary policy shock. The interest income tax τ_t^i does not enter the Taylor rule directly; it affects the economy through the household bond Euler equation (1.15), which links the pre-tax policy rate R_t set by the central bank to the after-tax return relevant for household saving decisions.

1.5 Equilibrium

Definition (Competitive Rational Expectations Equilibrium). Given exogenous shock processes $\{z_t, d_t, \mu_t\}_{t=0}^\infty$ and initial conditions $(k_{-1}, k_{-1}^\tau, s_{C,-1}, s_{G,-1}, b_{-1})$, a *competitive rational expectations equilibrium* is a collection of sequences of quantities

$$\{c_t, i_t, g_t, x_t, y_t, h_t, u_t, k_t, k_t^\tau, b_t, \text{div}_t, s_{C,t}, s_{G,t}, x_{c,t}, x_{g,t}\}_{t=0}^\infty,$$

prices and shadow values

$$\{w_t, r_t^k, R_t, \pi_t, \pi_t^w, mc_t, q_t, \lambda_t, v_{\tau,t}, \tilde{w}_t, \tilde{v}_t^c, \tilde{v}_t^g, \tilde{v}_t^i\}_{t=0}^\infty,$$

and fiscal instrument sequences $\{\tau_t^c, \tau_t^w, \tau_t^k, \tau_t^i, \tau_t^d, \tau_t^{ic}, e_t^\tau\}_{t=0}^\infty$ such that:

- (i) Taking prices and tax rates as given, the representative household maximizes (1.1) subject to the budget constraint (1.6), capital accumulation (1.7), and tax basis accumulation (1.8), yielding the first-order conditions (1.9)–(1.15).

- (ii) Taking wages w_t and rental rates r_t^k as given, each intermediate goods firm $i \in [0, 1]$ minimizes costs, yielding factor demands (1.20)–(1.21), and sets its nominal price to maximize the present discounted value of dividends subject to Rotemberg adjustment costs and the deep habits demand structure (1.16)–(1.17), yielding customer-base shadow values (1.23)–(1.25) and the price Phillips curve (1.26).
- (iii) Each household sets its nominal wage to maximize utility subject to Rotemberg wage adjustment costs, yielding the wage Phillips curve (1.28).
- (iv) The government satisfies its budget constraint (1.31) with tax revenue defined by (1.32), and each fiscal instrument follows its policy rule (1.33)–(1.41).
- (v) The central bank sets the gross nominal interest rate R_t according to the Taylor rule (1.42).
- (vi) Labor, capital, bond, and goods markets clear according to conditions (1.43)–(1.45).
- (vii) Expectations $E_t[\cdot]$ are rational: agents' beliefs are consistent with the probability distribution implied by the model.

1.5.1 Market Clearing

In a symmetric equilibrium all intermediate goods firms make identical factor choices, so labor and capital markets clear when household factor supply equals aggregate firm demand:

$$h_t = \int_0^1 h_{i,t} di \quad (1.43)$$

$$k_{t-1} = \int_0^1 k_{i,t-1} di \quad (1.44)$$

In symmetric equilibrium these reduce to $h_{i,t} = h_t$ and $k_{i,t-1} = k_{t-1}$ for all $i \in [0, 1]$.

The bond market clears when household bond holdings equal the government's outstanding debt. Since both the household budget constraint (1.6) and the government budget constraint (1.31) use the same variable b_t , this condition holds by construction.

Goods market clearing requires total net output to equal total absorption. Combining the output definition (1.22) with the resource constraint (1.30) yields:

$$y_t = c_t + g_t + i_t + \frac{\gamma_1}{1 + \sigma_u} (u_t^{1+\sigma_u} - 1) k_{t-1} + \frac{\alpha_W}{2} (\pi_t^w - \bar{\pi})^2 w_t \quad (1.45)$$

Output y_t is already net of the Rotemberg price adjustment cost (see equation (1.22)), so only the utilization cost and wage adjustment cost appear on the right-hand side. By Walras' law, the bond market clearing condition is redundant given (1.43), (1.44), and (1.45).

1.6 Solving the model

Probably best to use some linearization method. We used the AIM algorithm (Anderson and Moore (1985)) in Willis' class and I still have the code for that. Alternatively, we used Christiano's method of undetermined coefficients (Christiano (2002)) in Corbae's class. Chris Sims has the code for his method (Sims (2002)) available on his website. Uhlig (1998) does a good job summarizing these methods.

1.6.1 Characterizing equations

Altogether, this model has 36 endogenous variables and 28 exogenous parameters. The 36 endogenous variables are characterized by the 37 equations listed in the sections below. By Walras' Law, one of the market clearing conditions is redundant. In the computation, we discard the resource constraint (T.4.11).

$$\begin{aligned}
\text{Endogenous variables: } & \left\{ c_t, \lambda_t, l_t, i_t, b_t, v_t, k_t, k_t^\tau, \varepsilon_t^b \right\} \\
& \left\{ Y_t, y_{i,t}^d, P_t, \lambda_{p,t} \right\} \\
& \left\{ y_{i,t}, \tilde{k}_{i,t}, d_{i,t}, r_t^k, w_t, p_{i,t}, z_t \right\} \\
& \{ G_t, X_t \} \\
& \{ r_t \} \\
& \{ p_t, d_t, x_t, g_t, k_{i,t}, l_{i,t}, C_t, L_t, B_t, K_t, K_t^\tau, D_t, I_t \} \\
\text{Exogenous parameters: } & \left\{ \tau_t^c, \tau_t^i, \tau_t^l, \tau_t^k, \tau_t^{ic}, \tau_t^d, \delta_t^\tau, e_t^\tau \right\} \\
& \{ \beta, \zeta, \sigma \} \\
& \{ \delta_0, \delta_1, \delta_2, \gamma \} \\
& \{ \rho_\varepsilon, \mu_\varepsilon, \sigma_\varepsilon \} \\
& \{ \rho_\lambda, \mu_\lambda, \sigma_\lambda \} \\
& \{ \alpha, \theta \} \\
& \{ \gamma_{x,t} \} \\
& \{ \beta_r, \phi_1, \phi_2, \rho_r \}
\end{aligned}$$

There are 19 unknown variables (not counting those from the intermediate goods producers and the shocks, $\varepsilon_t^b, \lambda_{p,t}$ - for which we will have equations that we haven't specified yet) : $i, k, b, c, k^\tau, \lambda, g, x, d, P, w, z, \varepsilon^b, r, r^k, v, l, Y, M$. These breakdown into the following groups:

- 4 choice variables (Household): i, v, l, c
- 4 prices: P, w, r, r^k
- 2 choice variables (gov't): g, x , reallyly $2 + n$ where n is the number tax instruments
- 1 choice variable (monetary authority): M
- 2 exogenous state: z, ε
- 5 endogenous state variable: k, b, d, k^τ, Y

Solve for these unknowns with the following equations:

- 5 FOCs, household $\implies i, v, l, c, b$
- 3 from FOCs, intermediate goods producers $\implies r^k, w, P$ (These FOC's give demand for effective capital and labor. So together with the HH supply of these factors we can use our market clearing conditions to find the eq'm factor prices.)
- 1 from the profit function of intermediate goods producers $\implies d$

- 1 from FOCs, final goods producer $\implies Y$
- 1 from the zero profit condition for the final goods producer $\implies P$ (This together with the FOC's from the intermediate goods producers gives you the aggregate price index.)
- 1 Gov't budget constraint (+ n fiscal policy rules) $\implies X$, (+ n tax rates)
- 1 Resource constraint $\implies G$
- 2 laws of motion $\implies k^\tau, k$
- 3 exog shock process $\implies z, \varepsilon^b, \lambda_p$
- 1 Taylor rule for monetary authority $\implies r, M$ (There is the Taylor Rule plus the fact that $r = \max(M, 0)$ that determines both these.)

1.7 Calibration and Estimation

The model's parameter vector is partitioned into two groups. A subset of parameters is *calibrated*: fixed prior to estimation at values drawn from the literature or from long-run data targets, because they are weakly identified by the business-cycle observables used in estimation. The remaining parameters are *estimated* via Bayesian methods using U.S. quarterly data, following Zubairy (2010) and Smets and Wouters (2003). We use Dynare to compute the posterior mode via a Monte Carlo-based optimization routine and to characterize the posterior distribution via the Metropolis-Hastings algorithm with 1,500,000 draws across two chains, discarding the first one-third of draws as burn-in.

1.7.1 Calibrated Parameters

Table 1 reports the calibrated parameters. The discount factor β is set to imply a 3% annual real interest rate. The quarterly economic depreciation rate $\delta = 0.025$ corresponds to a standard 10% annual rate. The capital income share $\theta = 0.30$ matches the average labor share of income in U.S. national accounts. The price elasticity of substitution $\eta = 5.3$ implies a steady-state markup of roughly 23%, while the labor variety elasticity $\eta_w = 21.0$ is calibrated to match a 5% steady-state wage markup consistent with the Rotemberg wage-setting framework.

Steady-state hours worked $\bar{h} = 0.50$ normalizes the time endowment to unity and follows Zubairy (2010). The government spending share $\bar{g}/\bar{y} = 0.18$ and debt-to-output ratio $\bar{b}/\bar{y} = 0.33$ match U.S. historical averages. Steady-state inflation is normalized to $\bar{\pi} = 1$ (zero net inflation).

The steady-state tax rates are set to match average effective tax rates for the U.S. economy. The capital income tax $\bar{\tau}^k = 0.41$ and labor income tax $\bar{\tau}^w = 0.23$ follow Zubairy (2010). The consumption tax $\bar{\tau}^c = 0.10$ approximates the combined federal excise and average state sales tax burden. The interest income tax rate $\bar{\tau}^i = 0.20$ and dividend tax rate $\bar{\tau}^d = 0.15$ reflect average effective rates on capital income components. The investment tax credit and immediate expensing rates are calibrated to zero in steady state ($\bar{\tau}^{ic} = \bar{e}^\tau = 0$), reflecting the baseline tax code without active bonus depreciation or ITC provisions; both are active as stochastic fiscal instruments. The tax depreciation rate $\delta^\tau = 0.025$ equals the economic depreciation rate as a baseline. The dividend share of output $\bar{\text{div}}/\bar{y} = 0.11$ is

calibrated to match a realistic profit-to-output ratio consistent with U.S. data; the fixed operating cost ψ in the production function is determined residually to hit this target in steady state.

Table 1: Calibrated Parameters

Parameter	Description	Value
<i>Preferences and technology</i>		
β	Discount factor	$1/(1.03)^{1/4}$
δ	Quarterly economic depreciation rate	0.025
θ	Capital income share	0.30
η	Elasticity of substitution, goods varieties	5.3
η_w	Elasticity of substitution, labor varieties	21.0
<i>Steady-state targets</i>		
$\bar{\pi}$	Steady-state gross inflation	1.0
$\bar{h}$	Steady-state hours worked	0.50
$\bar{g}/\bar{y}$	Government spending share of output	0.18
$\bar{b}/\bar{y}$	Debt-to-output ratio	0.33
$\bar{\text{div}}/\bar{y}$	Dividend share of output	0.11
<i>Steady-state tax rates</i>		
$\bar{\tau}^k$	Capital income tax rate	0.41
$\bar{\tau}^w$	Labor income tax rate	0.23
$\bar{\tau}^c$	Consumption tax rate	0.10
$\bar{\tau}^i$	Interest income tax rate	0.20
$\bar{\tau}^d$	Dividend tax rate	0.15
$\bar{\tau}^{ic}$	Investment tax credit rate	0.00
$\bar{e}^\tau$	Immediate expensing rate	0.00
δ^τ	Tax depreciation rate	0.025

1.7.2 Bayesian Estimation

The remaining 57 parameters are estimated jointly using Bayesian techniques. We match 12 quarterly observables from U.S. data to 12 of the model's measurement equations:

1. $\hat{c}_t = 100(c_t - \bar{c})/\bar{c}$ — private consumption (percent deviation from SS)
2. $\hat{i}_t = 100(i_t - \bar{i})/\bar{i}$ — private investment (percent deviation)
3. $\hat{g}_t = 100(g_t - \bar{g})/\bar{g}$ — government purchases (percent deviation)
4. $\pi_t - \bar{\pi}$ — gross price inflation (level deviation)
5. $R_t - \bar{R}$ — gross nominal interest rate (level deviation)
6. $(\tau_t^k - \bar{\tau}^k)/\bar{\tau}^k$ — capital income tax rate (fractional deviation)
7. $(\tau_t^w - \bar{\tau}^w)/\bar{\tau}^w$ — labor income tax rate (fractional deviation)
8. $100(b_t - \bar{b})/\bar{b}$ — government debt (percent deviation)
9. $(\tau_t^c - \bar{\tau}^c)/\bar{\tau}^c$ — consumption tax rate (fractional deviation)
10. $(\tau_t^d - \bar{\tau}^d)/\bar{\tau}^d$ — dividend tax rate (fractional deviation)
11. $\tau_t^{ic} - \bar{\tau}^{ic}$ — investment tax credit rate (level deviation)

12. $e_t^\tau - \bar{e}^\tau$ — expensing rate (level deviation)

Transfer payments x_t are excluded from the observable set because of well-known data quality concerns. The interest income tax rate τ_t^i has no direct observable; its shock ε_t^i is identified indirectly through the bond Euler equation and debt dynamics. The model therefore has 13 structural shocks and 12 observables, with ε_t^i being the over-identified shock. Observed data are demeaned prior to estimation (`prefilter` = 1) to accommodate potential mismatches between model steady states and sample means.

Table 2 reports the estimated parameters along with their prior distributions. Prior choices closely follow Zubairy (2010) for the 28 parameters common to both models. Deep habit parameters b_c , θ^c , b_g , θ^g receive Beta priors on $(0, 1)$ consistent with their interpretation as persistence parameters. The risk aversion coefficient γ , utilization curvature σ_u , investment adjustment cost κ , and nominal rigidity parameters α_P , α_W receive Normal priors centered near values that generate moderate nominal rigidity. Fiscal rule AR coefficients ρ_j all receive Beta(0.7, 0.2) priors reflecting high but uncertain persistence. Debt-feedback coefficients for tax rules receive Gamma priors (positive), while those for spending rules receive Normal priors centered below zero (negative feedback). Output-feedback coefficients receive Normal priors centered near zero. All shock standard deviations receive inverse-Gamma priors. The 15 parameters for the five extension fiscal rules (ρ_c , ρ_{ic} , ρ_{ti} , ρ_{td} , $\rho_{e\tau}$ and their debt and output responses) and the cross-shock correlation ϕ_{wd} are new to this paper; their priors are set symmetrically with the analogous Zubairy parameters.

2 Short Run Multipliers

1. FIGURES: impulse response functions for tax cuts (capital, income, consumption, investment) and gov't spending increase (% changes in output, emp, cons, inv, int rates, inflation) to policy change
2. TABLE: table with size of multipliers

3 Long Run Multipliers

FIGURE: graph of multiplier from each of fiscal policy measures over time

4 Sensitivity to Government Financing

FIGURE: graph of multiplier over time given how gov't financing temporary tax cuts (with higher levels of debt going forward or with paying back by increasing taxes). Maybe 3D graph with multiplier, years from cut, and year to pay back debt as axes.

5 Sensitivity to Monetary Policy

Talk about multipliers at and away from the zero-bound. What happens if hold interest rates constant?

6 Dead-weight Loss and Fiscal Policy

1. TABLE: table of DWL for each of policy responses
2. FIGURE: graph of DWL over time for policy response

Do some revenue neutral changes in tax policy and calculate welfare....

7 Sensitivity to Parameters

How do results change with change to key parameters...

8 Discussion

Note that hard to know preferences for gov't spending, but imagine that people know better what want, cutting taxes may be politically easier than increasing spending. Note other weaknesses of the modeling assumptions.

If consumption taxes are useful stabilization tool, and not just a tool for more revenue, countries like the US that don't have them should consider implementing them. This is especially true if the zero bound is a concern, since consumption tax cuts can be effective even when interest rates are at the zero bound. Just like a non-zero inflation target can help give monetary policy more room to maneuver, a non-zero consumption tax can give fiscal policy more room to maneuver.

9 Conclusion

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TECHNICAL APPENDIX

T-1 Derivation of household's first order conditions

FOC's (to go in appendix):

$$\frac{\partial U}{\partial c_t} : u_c(c_t, l_t, g_t) - \lambda_t(1 + \tau_t^c) = 0 \quad (\text{T.1.1})$$

$$\frac{\partial U}{\partial l_t} : u_l(c_t, l_t, g_t) + u_c(c_t, l_t, g_t) \frac{w_t(1 - \tau_t^l)}{P_t(1 + \tau_t^c)} = 0 \quad (\text{T.1.2})$$

$$\begin{aligned} \frac{\partial U}{\partial i_t} : & u_c(c_t, l_t, g_t) \frac{(-1 + \tau_t^{ic} + \tau_t^k e_t^\tau)}{(1 + \tau_t^c)} + \\ & \beta E_t u_c(c_{t+1}, l_{t+1}, g_{t+1}) \frac{1}{(1 + \tau_{t+1}^c)} \left[\frac{(1 - \tau_{t+1}^k) r_{t+1}^k v_{t+1}}{P_{t+1}} \left(1 - S \left(\frac{i_t}{i_{t-1}} \right) - i_t \left(\frac{S' \frac{i_t}{i_{t-1}}}{\frac{i_t}{i_{t-1}}} \right) \right) + \tau_{t+1}^k \delta_{t+1}^\tau (1 - e_t^\tau) \right] \\ & \beta^2 E_t \left[u_c(c_{t+2}, l_{t+2}, g_{t+2}) \left(\frac{1 - \tau_{t+2}^k}{1 + \tau_{t+2}^c} \right) \frac{r_{t+2}^k v_{t+2}}{P_{t+2}} \gamma \left(\frac{i_{t+1}}{i_t} - 1 \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right] = 0 \end{aligned} \quad (\text{T.1.3})$$

$$\frac{\partial U}{\partial b_t} : \frac{-u_c(c_t, l_t, g_t)}{P_t(1 + \tau_t^c)} + \beta E_t u_c(c_{t+1}, l_{t+1}, g_{t+1}) \left(\frac{(1 + r_{t+1}(1 - \tau_{t+1}^i))}{P_{t+1}(1 + \tau_{t+1}^c)} \right) = 0 \quad (\text{T.1.4})$$

$$\frac{\partial U}{\partial v_t} : u_c(c_t, l_t, g_t) \left(\frac{(1 - \tau_t^k) r_t^k k_{t-1}}{P_t(1 + \tau_t^c)} \right) - \beta E_t u_c(c_{t+1}, l_{t+1}, g_{t+1}) \left(\frac{(1 - \tau_{t+1}^k) r_{t+1}^k v_{t+1} \frac{\partial \delta(v_t)}{\partial v_t} k_{t-1}}{P_{t+1}(1 + \tau_{t+1}^c)} \right) = 0 \quad (\text{T.1.5})$$

T-1.0.1 Utility specification 0

This utility function is similar to that in Christiano et al. (2010), Traum and Yang (2010), and Zubairy (2010).

$$u(c_t, l_t, g_t) = \varepsilon_t^b \left(\frac{[c_t^\zeta (1 - l_t)^{(1-\zeta)}]^{1-\sigma}}{1 - \sigma} + \chi^g \frac{g_t^{1-\sigma_g}}{1 - \sigma_g} \right) \quad (\text{T.1.6})$$

in which the preference shock is assumed to follow the following AR(1) process:

$$\begin{aligned} \log(\varepsilon_t^b) &= \rho_\varepsilon \log(\varepsilon_{t-1}^b) + \eta_{\varepsilon,t} \\ \text{where } \eta_{\varepsilon,t} &\sim N(0, \sigma_\varepsilon) \quad \text{and} \quad \rho_\varepsilon \in [0, 1) \end{aligned} \quad (\text{T.1.7})$$

FOC's with this utility function are:

$$\frac{\partial U}{\partial c_t} : \frac{\varepsilon_t^b \zeta c_t^{\zeta(1-\sigma)-1} (1-l_t)^{(1-\zeta)(1-\sigma)}}{1+\tau_t^c} = \lambda_t \quad (\text{T.1.8})$$

$$\frac{\partial U}{\partial l_t} : \left(\frac{1-\zeta}{\zeta} \right) \left(\frac{c_t}{1-l_t} \right) = \frac{w_t(1-\tau_t^l)}{P_t(1+\tau_t^c)} \quad (\text{T.1.9})$$

$$\begin{aligned} \frac{\partial U}{\partial i_t} : & \varepsilon_t^b \left(\frac{c_t}{1-l_t} \right)^{(\zeta-1)(1-\sigma)} c_t^{-\sigma} \left(\frac{\tau_t^{ic} + \tau_t^k e_t^\tau - 1}{1+\tau_t^c} \right) + \dots \\ & \beta E_t \left[\varepsilon_{t+1}^b \left(\frac{c_{t+1}}{1-l_{t+1}} \right)^{(\zeta-1)(1-\sigma)} c_{t+1}^{-\sigma} \dots \right. \\ & \left. \left(\frac{[1-\tau_{t+1}^k] r_{t+1}^k v_{t+1}}{P_{t+1}(1+\tau_{t+1}^c)} \left[1 - \frac{\gamma}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 - \gamma \left(\frac{i_t}{i_{t-1}} \right) \left(\frac{i_t}{i_{t-1}} - 1 \right) \right] + \frac{\tau_{t+1}^k \delta_{t+1}^\tau (1-e_{t+1}^\tau)}{1+\tau_{t+1}^c} \right] + \dots \right. \\ & \left. \beta^2 E_t \left[\varepsilon_{t+2}^b \left(\frac{c_{t+2}}{1-l_{t+2}} \right)^{(\zeta-1)(1-\sigma)} c_{t+2}^{-\sigma} \frac{(1-\tau_{t+2}^k) r_{t+2}^k v_{t+2}}{P_{t+2}(1+\tau_{t+2}^c)} \gamma \left(\frac{i_{t+1}}{i_t} - 1 \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right] \right] = 0 \end{aligned} \quad (\text{T.1.10})$$

$$\begin{aligned} \frac{\partial U}{\partial b_t} : & \varepsilon_t^b \left(\frac{c_t}{1-l_t} \right)^{(\zeta-1)(1-\sigma)} c_t^{-\sigma} \frac{1}{P_t(1+\tau_t^c)} = \dots \\ & \beta E_t \left[\varepsilon_{t+1}^b \left(\frac{c_{t+1}}{1-l_{t+1}} \right)^{(\zeta-1)(1-\sigma)} c_{t+1}^{-\sigma} \frac{1+r_{t+1}(1-\tau_{t+1}^i)}{P_{t+1}(1+\tau_{t+1}^c)} \right] \end{aligned} \quad (\text{T.1.11})$$

$$\begin{aligned} \frac{\partial U}{\partial v_t} : & \varepsilon_t^b \left(\frac{c_t}{1-l_t} \right)^{(\zeta-1)(1-\sigma)} c_t^{-\sigma} \frac{(1-\tau_t^k) r_t^k k_{t-1}}{P_t(1+\tau_t^c)} = \dots \\ & \beta E_t \left[\varepsilon_{t+1}^b \left(\frac{c_{t+1}}{1-l_{t+1}} \right)^{(\zeta-1)(1-\sigma)} c_{t+1}^{-\sigma} \frac{(1-\tau_{t+1}^k) r_{t+1}^k v_{t+1}}{P_{t+1}(1+\tau_{t+1}^c)} (\delta_1 + \delta_2[v_t - 1]) k_{t-1} \right] \end{aligned} \quad (\text{T.1.12})$$

$$(\text{T.1.13})$$

T-1.0.2 Alternative Setup

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[u(c_t, l_t, g_t) - \lambda_t \left[(1+\tau_t^c) c_t + \frac{b_t}{P_t} + i_t - \frac{(1+r_t(1-\tau_t^i)) b_{t-1}}{P_t} - \frac{(1-\tau_t^l) w_t l_t}{P_t} - \frac{(1-\tau_t^k) r_t^k v_t k_{t-1}}{P_t} - \tau_t^k \delta \right] \right] \quad (\text{T.1.14})$$

FOCS:

$$\frac{\partial U}{\partial c_t} : u_c(c_t, l_t, g_t) - \lambda_t(1+\tau_t^c) = 0 \quad (\text{T.1.15})$$

$$\frac{\partial U}{\partial l_t} : u_l(c_t, l_t, g_t) + \frac{\lambda_t w_t (1-\tau_t^l)}{P_t} = 0 \quad (\text{T.1.16})$$

$$\frac{\partial U}{\partial b_t} : \frac{\lambda_t}{P_t} = \beta E_t \left[\frac{\lambda_{t+1}(1 + r_{t+1}(1 - \tau_{t+1}^i))}{P_{t+1}} \right] \quad (\text{T.1.17})$$

$$\frac{\partial U}{\partial v_t} : \frac{\lambda_t(1 - \tau_t^k)r_t^k k_{t-1}}{P_t} = Q_t(\delta_1 + \delta_2(v_t - 1))k_{t-1} \quad (\text{T.1.18})$$

$$\frac{\partial U}{\partial k_t} : Q_t = \beta E_t \left[\frac{\lambda_{t+1}(1 - \tau_{t+1}^k)r_{t+1}^k v_{t+1}}{P_{t+1}} + Q_{t+1}(1 - \delta(v_{t+1})) \right] \quad (\text{T.1.19})$$

$$\begin{aligned} \frac{\partial U}{\partial i_t} : \lambda_t(1 - \tau_t^{ic} - \tau_t^k e_t^\tau) = Q_t \left[1 - S \left(\frac{i_t}{i_{t-1}} \right) - S' \left(\frac{i_t}{i_{t-1}} \right) \frac{i_t}{i_{t-1}} \right] + \\ \beta E_t \left[\lambda_{t+1} \tau_{t+1}^k \delta_{t+1}^\tau (1 - e_t^\tau) + Q_{t+1} S' \left(\frac{i_{t+1}}{i_t} \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right] \end{aligned} \quad (\text{T.1.20})$$

T-1.0.3 HH FOC's in the SS

$$\frac{\partial U}{\partial c} : \frac{\zeta \bar{c}^{\zeta(1-\sigma)-1} (1 - \bar{l})^{(1-\zeta)(1-\sigma)}}{(1 + \tau^c)} = \bar{\lambda} \quad (\text{T.1.21})$$

$$\frac{\partial U}{\partial l} : \frac{(1 - \zeta)\bar{c}}{\zeta(1 - \bar{l})} = \frac{\bar{w}(1 - \tau^l)}{(1 + \tau^c)} \quad (\text{T.1.22})$$

$$\frac{\partial U}{\partial i} : (1 - \tau^{ic} - \tau^k e^\tau) = \beta \left[(1 - \tau^k) \bar{r}^k + \tau^k \delta^\tau (1 - e^\tau) \right] \quad (\text{T.1.23})$$

$$\frac{\partial U}{\partial b} : 1 = \beta(1 + \bar{r}(1 - \tau^i)) \quad (\text{T.1.24})$$

$$\frac{\partial U}{\partial v} : \bar{k} = \beta \delta_1 \quad (\text{T.1.25})$$

T-1.0.4 SS of HH constraints/laws of motion

- $\bar{k} = (1 - \delta(\bar{v}))\bar{k} + \bar{i} \implies \bar{k}\delta(\bar{v}) = \bar{i}$, if $\bar{v} = 1$, then $\bar{i} = \bar{k}\delta_0$
- $\delta(\bar{v}) = \delta_0 + \delta_1(\bar{v} - 1) + \frac{\delta_2}{2}(\bar{v} - 1)^2$, if $\bar{v} = 1$, then $\delta(\bar{v}) = \delta_0$
- $S(\bar{i}) = 0$
- $\bar{k}^\tau = (1 - \delta^\tau)\bar{k}^\tau + \bar{i}(1 - e^\tau) \implies \bar{k}^\tau \delta^\tau = \bar{i}(1 - e^\tau)$
- Seems as though can normalize $\bar{P} = 1$ and $\bar{v} = 1$. Problems with this??

T-1.0.5 SS of non-HH equations

- Resource constraint: $\bar{Y} = \bar{C} + \delta_0 \bar{K} + \bar{G}$
- Government Budget Constraint: $\tau^c \bar{C} + \tau^l \bar{L} \bar{w} - (\bar{r}(1 - \tau^i)) \bar{B} + \tau^k \bar{r}^k \bar{K} + \tau^d \bar{D} = \bar{G} + \bar{X} + \tau^{ic} \delta_0 \bar{K} + \tau^k \delta_0 \bar{K} (1 - e^\tau) + \tau^k e^\tau \delta_0 \bar{K}$ It might be that profits of the intermediate goods producers are zero in the SS- in which case, $\bar{D} = 0$.
- Technology shock: $\bar{z} = 1$
- SS conditions from Intermediate Goods Producers:
 - marginal cost: $mc = \frac{\bar{w}(1-\alpha)\bar{r}^{k\alpha}}{\alpha^\alpha(1-\alpha)^{(1-\alpha)}}$
 - labor-capital ratio: $\frac{\bar{K}}{\bar{L}} = \left(\frac{\alpha}{1-\alpha}\right) \frac{\bar{w}}{\bar{r}^k}$
 - output: $\bar{Y} = \bar{K}^\alpha \bar{L}^{(1-\alpha)}$
- Final Goods Producers: $\bar{P} = 1$ (*I think*).

T-1.0.6 Steady state values of state variables:

From the Calvo pricing equation, we can normalize $\bar{P}$:

$$\bar{P} = 1 \tag{T.1.26}$$

We can calibrate δ_0 such that we can normalize our SS capital utilization:

$$\bar{V} = 1 \tag{T.1.27}$$

We can normalize the mean of the TFP process (and know that the disturbance term = 0 in the SS):

$$\bar{Z} = 1 \tag{T.1.28}$$

We can normalize the mean of the taste shock (and know that the disturbance term = 0 in the SS):

$$\bar{\varepsilon}^b = 1 \tag{T.1.29}$$

We know that the disturbance term to the price markup is zero in the SS, thus:

$$\bar{\lambda}_p = \lambda_p \tag{T.1.30}$$

We can solve for the SS interest rate directly from the HH FOC for demand for gov't bonds (or from the Taylor Rule):

$$\bar{r} = \frac{1 - \beta}{\beta(1 - \tau^i)} \tag{T.1.31}$$

The FOC for the HH investment decision, we can find the SS return on capital:

$$\bar{r}^k = \frac{\left(\frac{1 - \tau^{ic} - \tau^k e^\tau}{\beta}\right) - \tau^k \delta^\tau (1 - e^\tau)}{(1 - \tau^k)} \tag{T.1.32}$$

The FOC for labor demand from the int goods producer's problem gives the SS wage rate:

$$\bar{w} = (1 - \alpha) \left(\frac{\bar{r}^k}{\alpha} \right)^{\frac{-\alpha}{1-\alpha}} \quad (\text{T.1.33})$$

Calibration gives the SS labor supply (I think this implies the Frisch elasticity of labor paramter = 1 – see Zubairy (2010)):

$$\bar{L} = 0.5 \quad (\text{T.1.34})$$

The FOC for effective capital demand from the int goods producer's problem gives the SS level of capital:

$$\bar{K} = \left(\frac{\alpha}{1 - \alpha} \right) \frac{\bar{w} \bar{L}}{\bar{r}^k} \quad (\text{T.1.35})$$

The law of motion for the tax basis of the capital stock gives the SS value for the tax basis of the capital stock:

$$\bar{K}^\tau = \frac{\delta_0 \bar{K} (1 - e^\tau)}{\delta^\tau} \quad (\text{T.1.36})$$

The law of motion for the capital stock gives SS investment:

$$\bar{I} = \delta_0 \bar{K} \quad (\text{T.1.37})$$

The int goods producers' production function gives the aggregate level of output in the SS:

$$\bar{Y} = \bar{K}^\alpha \bar{L}^{(1-\alpha)} \quad (\text{T.1.38})$$

The int goods producers' profit function gives aggregate profits in the SS:

$$\bar{D} = \bar{Y} - \bar{w} \bar{L} - \bar{r}^k \bar{K} \quad (\text{T.1.39})$$

The HH FOC for labor gives SS consumption:

$$\bar{C} = \frac{\bar{w}(1 - \tau^l)(1 - \bar{L})\zeta}{(1 - \zeta)(1 + \tau^c)} \quad (\text{T.1.40})$$

The HH FOC for consumption gives SS marginal utility of consumption (the shadow price of consumption):

$$\bar{\lambda} = \frac{\zeta \bar{c}^{\zeta(1-\sigma)-1} (1 - \bar{l})^{(1-\zeta)(1-\sigma)}}{(1 + \tau^c)} \quad (\text{T.1.41})$$

The resource constraint gives SS government spending:

$$\bar{G} = \bar{Y} - \bar{C} - \bar{I} \quad (\text{T.1.42})$$

The gov't budget constraint gives SS bond issuance necessary to finance $\bar{G}$:

$$\bar{B} = \frac{[\tau^c \bar{C} + \tau^l \bar{L} \bar{w} + \tau^k \bar{r}^k \bar{K} + \tau^d \bar{D} - \tau^{ic} \bar{I} - \tau^k \delta^\tau \bar{K}^\tau - \tau^k e^\tau \bar{I} - (1 - \gamma_x) \bar{G}]}{\bar{r}(1 - \tau^i)} \quad (\text{T.1.43})$$

The assumption regarding what fraction of the federal budget goes to X gives SS government transfers:

$$\bar{X} = \gamma_x * \left[\tau^c \bar{C} + \tau^l \bar{L} \bar{w} + \bar{B} + \tau^k \bar{r}^k \bar{K} + \tau^d \bar{D} - \tau^{ic} \bar{I} - \tau^k \delta^\tau \bar{K}^\tau - \tau^k e^\tau \bar{I} - \bar{B} [1 + \bar{r}(1 - \tau^i)] \right] \quad (\text{T.1.44})$$

T-2 Derivation of monopolistic competition intermediate goods demand and aggregate price Equations

Maybe repeated below, but pasting to temporarily have in one place and preserve:

The FOC's are:

$$p_t(1 + \lambda_{p,t}) \left(\int_0^1 y_{i,t}^{\left(\frac{1}{1+\lambda_{p,t}}\right)} di \right)^{\lambda_{p,t}} \left(\frac{1}{1 + \lambda_{p,t}} \right) y_{i,t}^{\frac{1}{1+\lambda_{p,t}}-1} - p_{i,t} = 0, \forall i \quad (\text{T.2.1})$$

Dividing the FOCs for intermediate inputs i and $j \implies$

$$\frac{p_{i,t}}{p_{j,t}} = \left(\frac{y_{i,t}}{y_{j,t}} \right)^{\frac{1}{1+\lambda_{p,t}}-1} = \left(\frac{y_{i,t}}{y_{j,t}} \right)^{\frac{-\lambda_{p,t}}{1+\lambda_{p,t}}} \quad (\text{T.2.2})$$

Rearranging:

$$\begin{aligned} \implies p_{i,t} &= \left(\frac{y_{j,t}}{y_{i,t}} \right)^{\frac{\lambda_{p,t}}{1+\lambda_{p,t}}} p_{j,t} \\ \implies p_{i,t} &= y_{j,t}^{\frac{\lambda_{p,t}}{1+\lambda_{p,t}}} p_{j,t} y_{i,t}^{\frac{-\lambda_{p,t}}{1+\lambda_{p,t}}} \\ \implies p_{i,t} y_{i,t} &= p_{j,t} y_{j,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} y_{i,t}^{\frac{1}{1+\lambda_{p,t}}} \end{aligned} \quad (\text{T.2.3})$$

Then integrate this condition to yield:

$$\begin{aligned} \int_0^1 p_{i,t} y_{i,t} di &= p_{j,t} y_{j,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} \int_0^1 y_{i,t}^{\frac{1}{1+\lambda_{p,t}}} di \\ &= p_{j,t} y_{j,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} (Y_t)^{\frac{1}{1+\lambda_{p,t}}} \end{aligned} \quad (\text{T.2.4})$$

The zero profit condition implies $P_t Y_t = \int_0^1 p_{i,t} y_{i,t} di$. Plugging this into the above and we find:

$$\begin{aligned} P_t Y_t &= p_{j,t} y_{j,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} (Y_t)^{\frac{1}{1+\lambda_{p,t}}} \\ \implies P_t Y_t &= p_{j,t} y_{j,t}^{\frac{\lambda_{p,t}}{1+\lambda_{p,t}}} (Y_t)^{\frac{1}{1+\lambda_{p,t}}} \\ \implies P_t &= p_{j,t} y_{j,t}^{\frac{\lambda_{p,t}}{1+\lambda_{p,t}}} (Y_t)^{\frac{-\lambda_{p,t}}{1+\lambda_{p,t}}} \end{aligned} \quad (\text{T.2.5})$$

Which implies the demand function (just rearranging terms):

$$y_{i,t} = \left(\frac{p_{i,t}}{P_t} \right)^{\frac{-(1+\lambda_{p,t})}{\lambda_{p,t}}} Y_t, \forall i \quad (\text{T.2.6})$$

To find P_t use the zero profit condition:

$$\begin{aligned}
P_t Y_t &= \int_0^1 p_{i,t} y_{i,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} (Y_t)^{\frac{1}{1+\lambda_{p,t}}} di \\
\Rightarrow P_t Y_t &= \int_0^1 p_{i,t} \left(\left(\frac{p_{i,t}}{P_t} \right)^{\frac{-(1+\lambda_{p,t})}{\lambda_{p,t}}} Y_t \right)^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} (Y_t)^{\frac{1}{1+\lambda_{p,t}}} di \\
\Rightarrow P_t Y_t &= \int_0^1 p_{i,t} \left(\frac{p_{i,t}}{P_t} \right)^{\frac{-(1+\lambda_{p,t})}{\lambda_{p,t}}} Y_t di \\
\Rightarrow P_t Y_t &= \int_0^1 p_{i,t} p_{i,t}^{\frac{-(1+\lambda_{p,t})}{\lambda_{p,t}}} P_t^{\frac{(1+\lambda_{p,t})}{\lambda_{p,t}}} Y_t di \\
\Rightarrow P_t Y_t &= \int_0^1 p_{i,t}^{\frac{-1}{\lambda_{p,t}}} di Y_t P_t^{\frac{1+\lambda_{p,t}}{\lambda_{p,t}}} \\
\Rightarrow P_t &= \int_0^1 p_{i,t}^{\frac{-1}{\lambda_{p,t}}} di P_t^{\frac{1+\lambda_{p,t}}{\lambda_{p,t}}} \\
\Rightarrow P_t^{\frac{-1}{\lambda_{p,t}}} &= \int_0^1 p_{i,t}^{\frac{-1}{\lambda_{p,t}}} di \\
\Rightarrow P_t &= \left(\int_0^1 p_{i,t}^{\frac{-1}{\lambda_{p,t}}} di \right)^{-\lambda_{p,t}}
\end{aligned} \tag{T.2.7}$$

The profit maximization approach to solving for each intermediate good demand equation $y_{i,t}$ and the aggregate price equation P_t is the following.

$$\max_{y_{i,t}} P_t Y_t^d - \int_0^1 p_{i,t} y_{i,t} di \tag{T.2.8}$$

The FOC's are:

$$p_t(1 + \lambda_{p,t}) \left(\int_0^1 y_{i,t}^{\left(\frac{1}{1+\lambda_{p,t}}\right)} di \right)^{\lambda_{p,t}} \left(\frac{1}{1 + \lambda_{p,t}} \right) y_{i,t}^{\frac{1}{1+\lambda_{p,t}}-1} - p_{i,t} = 0, \forall i \tag{T.2.9}$$

Dividing the FOCs for intermediate inputs i and $j \Rightarrow$

$$\frac{p_{i,t}}{p_{j,t}} = \left(\frac{y_{i,t}}{y_{j,t}} \right)^{\frac{1}{1+\lambda_{p,t}}-1} = \left(\frac{y_{i,t}}{y_{j,t}} \right)^{\frac{-\lambda_{p,t}}{1+\lambda_{p,t}}} \tag{T.2.10}$$

Rearranging:

$$\begin{aligned}
\Rightarrow p_{i,t} &= \left(\frac{y_{j,t}}{y_{i,t}} \right)^{\frac{\lambda_{p,t}}{1+\lambda_{p,t}}} p_{j,t} \\
\Rightarrow p_{i,t} &= y_{j,t}^{\frac{\lambda_{p,t}}{1+\lambda_{p,t}}} p_{j,t} y_{i,t}^{\frac{-\lambda_{p,t}}{1+\lambda_{p,t}}} \\
\Rightarrow p_{i,t} y_{i,t} &= p_{j,t} y_{j,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} y_{i,t}^{\frac{1}{1+\lambda_{p,t}}}
\end{aligned} \tag{T.2.11}$$

Then integrate this condition to yield:

$$\begin{aligned}\int_0^1 p_{i,t} y_{i,t} di &= p_{j,t} y_{j,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} \int_0^1 y_{i,t}^{\frac{1}{1+\lambda_{p,t}}} di \\ &= p_{j,t} y_{j,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} (Y_t)^{\frac{1}{1+\lambda_{p,t}}}\end{aligned}\quad (\text{T.2.12})$$

The zero profit condition implies $P_t Y_t = \int_0^1 p_{i,t} y_{i,t} di$. Plugging this into the above and we find:

$$\begin{aligned}P_t Y_t &= p_{j,t} y_{j,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} (Y_t)^{\frac{1}{1+\lambda_{p,t}}} \\ \implies P_t Y_t &= p_{j,t} y_{j,t}^{\frac{\lambda_{p,t}}{1+\lambda_{p,t}}} (Y_t)^{\frac{1}{1+\lambda_{p,t}}} \\ \implies P_t &= p_{j,t} y_{j,t}^{\frac{\lambda_{p,t}}{1+\lambda_{p,t}}} (Y_t)^{\frac{-\lambda_{p,t}}{1+\lambda_{p,t}}}\end{aligned}\quad (\text{T.2.13})$$

Which implies the demand function (just rearranging terms):

$$y_{i,t} = \left(\frac{p_{i,t}}{P_t}\right)^{\frac{-(1+\lambda_{p,t})}{\lambda_{p,t}}} Y_t, \forall i \quad (\text{T.2.14})$$

To find P_t use the zero profit condition:

$$\begin{aligned}P_t Y_t &= \int_0^1 p_{i,t} y_{i,t}^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} (Y_t)^{\frac{1}{1+\lambda_{p,t}}} di \\ \implies P_t Y_t &= \int_0^1 p_{i,t} \left(\left(\frac{p_{i,t}}{P_t}\right)^{\frac{-(1+\lambda_{p,t})}{\lambda_{p,t}}} Y_t \right)^{\left(\frac{\lambda_{p,t}}{1+\lambda_{p,t}}\right)} (Y_t)^{\frac{1}{1+\lambda_{p,t}}} di \\ \implies P_t Y_t &= \int_0^1 p_{i,t} \left(\frac{p_{i,t}}{P_t}\right)^{\frac{-(1+\lambda_{p,t})}{\lambda_{p,t}}} Y_t di \\ \implies P_t Y_t &= \int_0^1 p_{i,t} p_{i,t}^{\frac{-(1+\lambda_{p,t})}{\lambda_{p,t}}} P_t^{\frac{(1+\lambda_{p,t})}{\lambda_{p,t}}} Y_t di \\ \implies P_t Y_t &= \int_0^1 p_{i,t}^{\frac{-1}{\lambda_{p,t}}} di Y_t P_t^{\frac{1+\lambda_{p,t}}{\lambda_{p,t}}} \\ \implies P_t &= \int_0^1 p_{i,t}^{\frac{-1}{\lambda_{p,t}}} di P_t^{\frac{1+\lambda_{p,t}}{\lambda_{p,t}}} \\ \implies P_t^{\frac{-1}{\lambda_{p,t}}} &= \int_0^1 p_{i,t}^{\frac{-1}{\lambda_{p,t}}} di \\ \implies P_t &= \left(\int_0^1 p_{i,t}^{\frac{-1}{\lambda_{p,t}}} di \right)^{-\lambda_{p,t}}\end{aligned}\quad (\text{T.2.15})$$

The cost minimization approach to solving for each intermediate good demand equation $y_{i,t}$ and the aggregate price equation P_t is the following.

$$\min_{y_{i,t}} \int_0^1 p_{i,t} y_{i,t} di \quad \text{s.t.} \quad Y_t \leq \left(\int_0^1 y_{i,t}^{\frac{1}{1+\lambda_{p,t}}} di \right)^{1+\lambda_{p,t}} \quad (\text{T.2.16})$$

The Lagrangian for this minimization problem is the following,

$$\mathcal{L} = \int_0^1 p_{i,t} y_{i,t} di + \lambda_y \left[Y_t - \left(\int_0^1 y_{i,t}^{\frac{1}{1+\lambda_{p,t}}} di \right)^{1+\lambda_{p,t}} \right] \quad (\text{T.2.17})$$

in which the multiplier λ_y has the interpretation of being the marginal cost of an extra unit of aggregated output. That is, λ_y is the price of aggregate output P_t .

$$\mathcal{L} = \int_0^1 p_{i,t} y_{i,t} di + P_t \left[Y_t - \left(\int_0^1 y_{i,t}^{\frac{1}{1+\lambda_{p,t}}} di \right)^{1+\lambda_{p,t}} \right] \quad (\text{T.2.18})$$

We need to finish this....

Intermediate goods producers derivations: FOC's:

$$\frac{\partial V}{\partial \tilde{k}_{i,t}} : \frac{\alpha p_{i,t} y_{i,t}}{\tilde{k}_{i,t}} - r_t^k = 0 \quad (\text{T.2.19})$$

$$\frac{\partial V}{\partial l_{i,t}} : \frac{(1-\alpha) p_{i,t} y_{i,t}}{l_{i,t}} - w_t = 0 \quad (\text{T.2.20})$$

Dividing Equation [T.2.19](#) by Equation [T.2.20](#) and rearranging some terms, one gets:

$$\frac{\tilde{k}_{i,t}}{l_{i,t}} = \left(\frac{\alpha}{1-\alpha} \right) \frac{w_t}{r_t^k} \quad (\text{T.2.21})$$

Plugging this ratio into Equation ?? and setting it equal to 1, one can find the amount of labor needed to produce one unit of output in terms of w_t, r_t^k, z_t , and the parameter α . Rearranging [T.2.21](#) to put $\tilde{k}_{i,t}$ in terms of $l_{i,t}$, factor input prices, and parameters then writing the cost function $(r_{t+j}^k \tilde{k}_{i,t+j} + w_{t+j} l_{i,t+j})$ in terms of $l_{i,t}$, factor input prices, and parameters, one can substitute in the equation for $l_{i,t}$. This yields the marginal cost of production in real terms (this procedure is described in Fernandez-Villaverde and Rubio-Ramirez (2006)):

$$mc_t = \frac{w_t^{(1-\alpha)} (r_t^k)^\alpha}{p_t z_t \alpha^\alpha (1-\alpha)^{(1-\alpha)}} \quad (\text{T.2.22})$$

N.B., the marginal cost function, mc_t does not depend upon i . This is because all firms receive the same technology shock and face the same factor input prices.

The second stage, where firms optimally choose price, can be setup as a profit maximization problem, taking mc_t as given. Assuming an interior solution, the problem can be written as:

$$V = \max_{p_{i,t}} E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \frac{\lambda_{t+j}}{\lambda_t} \left[\left(\frac{p_{i,t}}{P_{t+j}} - mc_{t+j} \right) y_{i,t+j} \right] \right), \quad (\text{T.2.23})$$

subject to:

$$y_{i,t+j} = \left(\frac{p_{i,t}}{P_{t+j}} \right)^{-\left(\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}} \right)} Y_{t+j} \quad (\text{T.2.24})$$

where Y_{t+j} is the demand for final goods in period $t+j$. N.B. there is no $p_{i,t+j}$ because prices are fixed at their value chose in period t .

Rewriting the second-stage problem gives:

$$\begin{aligned} V &= \max_{p_{i,t}} E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \frac{\lambda_{t+j}}{\lambda_t} \left[\left(\frac{p_{i,t}}{P_{t+j}} - mc_{t+j} \right) y_{i,t+j} \right] \right) \\ \Rightarrow V &= \max_{p_{i,t}} E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \frac{\lambda_{t+j}}{\lambda_t} \left[\left(\frac{p_{i,t}}{P_{t+j}} - mc_{t+j} \right) \left(\frac{p_{i,t}}{P_{t+j}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} Y_{t+j} \right] \right) \\ \Rightarrow V &= \max_{p_{i,t}} E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \frac{\lambda_{t+j}}{\lambda_t} \left[\left(\frac{p_{i,t}}{P_{t+j}} \right)^{\frac{-1}{\lambda_{p,t+j}}} - \left(\frac{p_{i,t}}{P_{t+j}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} mc_{t+j} \right] Y_{t+j} \right) \\ \Rightarrow V &= \max_{p_{i,t}} E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \frac{\lambda_{t+j}}{\lambda_t} \left[\left(\prod_{s=1}^j \frac{1}{\pi_{t+s}} \frac{p_{i,t}}{P_t} \right)^{\frac{-1}{\lambda_{p,t+j}}} - \left(\prod_{s=1}^j \frac{1}{\pi_{t+s}} \frac{p_{i,t}}{P_t} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} mc_{t+j} \right] Y_{t+j} \right) \end{aligned} \quad (\text{T.2.25})$$

In the above, $\pi_t = \frac{P_t}{P_{t-1}}$.

The FOC for this problem (where the intermediate-goods producer is choosing $p_{i,t}$) is given by:

$$\begin{aligned} \frac{\partial V}{\partial p_{i,t}} : E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \frac{\lambda_{t+j}}{\lambda_t} \left[\frac{-1}{\lambda_{p,t+j}} \left(\prod_{s=1}^j \frac{1}{\pi_{t+s}} \frac{p_{i,t}}{P_t} \right)^{\frac{-1}{\lambda_{p,t+j}}} \left(\frac{1}{p_{i,t}} \right) + \left(\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}} \right) \left(\prod_{s=1}^j \frac{1}{\pi_{t+s}} \frac{p_{i,t}}{P_t} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} mc_{t+j} \right] Y_{t+j} \right) \\ \Rightarrow \frac{\partial V}{\partial p_{i,t}} : E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \frac{\lambda_{t+j}}{\lambda_t} \left(\prod_{s=1}^j \frac{1}{\pi_{t+s}} \frac{1}{p_{i,t}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} \left[\frac{-1}{\lambda_{p,t+j}} \left(\prod_{s=1}^j \frac{1}{\pi_{t+s}} \frac{p_{i,t}}{P_t} \right) + \left(\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}} \right) mc_{t+j} \right] Y_{t+j} \right) = 0 \end{aligned} \quad (\text{T.2.26})$$

Rearranging:

$$\begin{aligned} \Rightarrow \frac{\partial V}{\partial p_{i,t}} : E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \frac{\lambda_{t+j}}{\lambda_t} \left(\prod_{s=1}^j \frac{1}{\pi_{t+s}} \frac{1}{p_{i,t}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} \left[\frac{1}{\lambda_{p,t+j}} \left(\prod_{s=1}^j \frac{1}{\pi_{t+s}} \frac{p_{i,t}}{P_t} \right) \right] Y_{t+j} \right) \\ = E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \frac{\lambda_{t+j}}{\lambda_t} \left(\prod_{s=1}^j \frac{1}{\pi_{t+s}} \frac{1}{p_{i,t}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} \left[\left(\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}} \right) mc_{t+j} \right] Y_{t+j} \right) \end{aligned}$$

(T.2.27)

Or (using notation without gross inflation):

$$\Rightarrow \frac{\partial V}{\partial p_{i,t}} : E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \left(\frac{p_{i,t}}{P_{t+j}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} \frac{Y_{t+j} \lambda_{t+j}}{\lambda_{p,t+j} \lambda_t} \left[\frac{(1 + \lambda_{p,t+j}) m c_{t+j} P_{t+j} - p_{i,t}}{p_{i,t} P_{t+j}} \right] \right) = 0 \quad \forall i, t$$

(T.2.28)

Which implies:

$$E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \left(\frac{p_{i,t}}{P_{t+j}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} \frac{Y_{t+j} \lambda_{t+j}}{\lambda_{p,t+j} \lambda_t} \left[\frac{(1 + \lambda_{p,t+j}) m c_{t+j} P_{t+j}}{p_{i,t} P_{t+j}} \right] \right) =$$

$$E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \left(\frac{p_{i,t}}{P_{t+j}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} \frac{Y_{t+j} \lambda_{t+j}}{\lambda_{p,t+j} \lambda_t} \left[\frac{p_{i,t}}{p_{i,t} P_{t+j}} \right] \right)$$

(T.2.29)

Which can be written as:

$$p_{i,t}^* = \frac{E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \left(\frac{p_{i,t}}{P_{t+j}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} \frac{Y_{t+j} \lambda_{t+j}}{\lambda_{p,t+j} \lambda_t} [(1 + \lambda_{p,t+j}) m c_{t+j}] \right)}{E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \left(\frac{p_{i,t}}{P_{t+j}} \right)^{-\frac{1+\lambda_{p,t+j}}{\lambda_{p,t+j}}} \frac{Y_{t+j} \lambda_{t+j}}{\lambda_{p,t+j} \lambda_t} \right)}$$

(T.2.30)

Assume a symmetric equilibrium ($\Rightarrow p_{i,t}^* = p_t^*, \forall i$).

Calvo pricing (with zero indexation) implies $P_t^{\frac{-1}{\lambda_{p,t}}} = \theta P_{t-1}^{\frac{-1}{\lambda_{p,t}}} + (1 - \theta) p_t^{*\frac{-1}{\lambda_{p,t}}}$ (you can get this from the definition of the Calvo model and the aggregate price index from the final goods producers problem). One can divide this equation through by $P_t^{\frac{-1}{\lambda_{p,t}}}$ and get $1 = \theta \left(\frac{P_{t-1}}{P_t} \right)^{\frac{-1}{\lambda_{p,t}}} + (1 - \theta) \left(\frac{p_t^*}{P_t} \right)^{\frac{-1}{\lambda_{p,t}}}$. Using the gross inflation variable, π_t defined above, we can write: $\left(\frac{p_t^*}{P_t} \right) = \left(\frac{1 - \theta \pi_t^{\frac{1}{\lambda_{p,t}}}}{1 - \theta} \right)^{-\lambda_{p,t}}$.

Using this, we can write the equation for p_t^* as:

Which can be written as:

$$p_t^* = \frac{E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \left(\frac{1 - \theta \pi_{t+j}^{\frac{1}{\lambda_{p,t+j}}}}{1 - \theta} \right)^{1 + \lambda_{p,t+j}} \frac{Y_{t+j} \lambda_{t+j}}{\lambda_{p,t+j} \lambda_t} [(1 + \lambda_{p,t+j}) m c_{t+j}] \right)}{E_t \left(\sum_{j=0}^{\infty} \beta^j \theta^j \left(\frac{1 - \theta \pi_{t+j}^{\frac{1}{\lambda_{p,t+j}}}}{1 - \theta} \right)^{1 + \lambda_{p,t+j}} \frac{Y_{t+j} \lambda_{t+j}}{\lambda_{p,t+j} \lambda_t} \right)}$$

(T.2.31)

The way most people seem to solve this is to log-linearize the equation. This turns the infinite sum into a difference equation. In addition, the analysis is greatly simplified by assuming that gross inflation (π) equals 1 in the steady state. Assuming that gross inflation equals 1 in the SS, I am getting the following equation for the price chosen by firms who can change price in period t :

$$p_t^* = (1 + \lambda_p)\bar{m}c(1 - \beta\theta)E_t \left[\sum_{j=0}^{\infty} \beta^j \theta^j \left[\frac{mc_{t+j} - \bar{m}c}{\bar{m}c} + \frac{P_{t+j} - \bar{P}}{\bar{P}} \right] + \frac{\lambda_p}{(1 + \lambda_p)} \sum_{j=0}^{\infty} \beta^j \theta^j \left(\frac{\lambda_{p,t+j} - \lambda_p}{\lambda_p} \right) \right] \quad (\text{T.2.32})$$

T-3 Derivation of real marginal costs

Dividing the period profits equation (??) by aggregate prices P_t gives the following real total costs function.

$$tc_t = \frac{r_t^k}{P_t} \tilde{k}_{i,t} + \frac{w_t}{P_t} l_{i,t} \quad (\text{T.3.1})$$

Substituting in the expression for $\tilde{k}_{i,t}$ from the capital-labor ratio equation (??) gives real total costs in terms of α , w_t , P_t , and $l_{i,t}$.

$$tc_t = \frac{1}{1-\alpha} \left(\frac{w_t}{P_t} \right) l_{i,t} \quad (\text{T.3.2})$$

Because the intermediate goods production exhibits constant returns to scale, we can find the real marginal cost mc_t by finding the amount of labor necessary to produce 1 unit of output, and substituting that value into the real total costs equation (T.3.2).

$$z_t \tilde{k}_{i,t}^\alpha l_{i,t}^{1-\alpha} = 1 \quad (\text{T.3.3})$$

Next, substitute in the equation for capital $\tilde{k}_{i,t}$ in terms of labor from the capital-labor ratio equation (??).

$$z_t \left(\frac{\alpha}{1-\alpha} \left[\frac{w_t}{r_t^k} \right] l_{i,t} \right)^\alpha l_{i,t}^{1-\alpha} = 1 \quad \Rightarrow \quad l_{i,t} = \frac{1}{z_t} \left(\frac{\alpha}{1-\alpha} \left[\frac{w_t}{r_t^k} \right] \right)^{-\alpha} \quad (\text{T.3.4})$$

Lastly, substitute the amount of labor necessary to produce one unit of output from (T.3.4) into the expression for real total costs (T.3.2) to get real marginal costs.

$$mc_t = \frac{1}{1-\alpha} \left(\frac{w_t}{P_t} \right) \frac{1}{z_t} \left(\frac{\alpha}{1-\alpha} \left[\frac{w_t}{r_t^k} \right] \right)^{-\alpha} = \frac{w_t^{1-\alpha} (r_t^k)^\alpha}{z_t P_t \alpha^\alpha (1-\alpha)^{1-\alpha}} \quad (\text{T.3.5})$$

T-4 Model solution

So we have 19 unknowns and 19 equations. We should be able to plug this into Dynare and solve the model.

T-4.0.1 Market clearing and resource constraint

The following 14 market clearing conditions close the model. The last equation (T.4.11) is a resource constraint, and can be thought of as goods market clearing. By Walras' Law, one of the following conditions is redundant.

$$c_t = C_t \quad (\text{T.4.1})$$

$$\int_0^1 l_{i,t} di = l_t = L_t \quad (\text{T.4.2})$$

$$i_t = I_t \tag{T.4.3}$$

$$\int_0^1 \tilde{k}_{i,t} di = v_t k_t = V_t K_t \tag{T.4.4}$$

$$k_t^\tau = K_t^\tau \tag{T.4.5}$$

$$\left(\int_0^1 y_{i,t}^{\frac{1}{1+\lambda_{p,t}}} di \right) = Y_t^d = Y_t \tag{T.4.6}$$

$$\left(\int_0^1 d_{i,t} \right) = d_t = D_t \tag{T.4.7}$$

$$\left(\int_0^1 p_{i,t}^{\frac{-1}{\lambda_{p,t}}} di \right)^{-\lambda_{p,t}} = P_t \tag{T.4.8}$$

$$g_t = G_t \tag{T.4.9}$$

$$x_t = X_t \tag{T.4.10}$$

$$Y_t = C_t + I_t + G_t \tag{T.4.11}$$

Table 2: Estimated Parameters: Prior Distributions

Parameter	Description	Prior	Mean	Std
<i>Deep habits</i>				
b_c	Private consumption habit depth	Beta	0.70	0.10
θ^c	Private habit stock persistence	Beta	0.80	0.10
b_g	Government consumption habit depth	Beta	0.70	0.10
θ^g	Government habit stock persistence	Beta	0.80	0.10
<i>Structural parameters</i>				
γ	Risk aversion / inverse EIS	Normal	2.00	1.00
σ_u	Utilization cost curvature	Normal	2.50	0.50
κ	Investment adjustment cost	Normal	2.00	0.50
α_P	Rotemberg price adjustment cost	Normal	17.0	5.00
α_W	Rotemberg wage adjustment cost	Normal	100.0	30.0
<i>Monetary policy (Taylor rule)</i>				
α_R	Interest rate smoothing	Beta	0.80	0.20
α_π	Inflation response	Normal	1.60	0.20
α_Y	Output response	Normal	0.10	0.05
<i>Exogenous process persistence</i>				
ρ_z	Technology shock	Beta	0.70	0.20
ρ_d	Preference shock	Beta	0.70	0.20
ρ_μ	Investment efficiency shock	Beta	0.70	0.20
<i>Fiscal rule AR coefficients</i>				
ρ_k	Capital tax rule persistence	Beta	0.70	0.20
ρ_w	Labor tax rule persistence	Beta	0.70	0.20
ρ_g	Government spending rule persistence	Beta	0.80	0.20
ρ_{tr}	Transfer rule persistence	Beta	0.70	0.20
ρ_c	Consumption tax rule persistence	Beta	0.70	0.20
ρ_{ic}	ITC rule persistence	Beta	0.70	0.20
ρ_{ti}	Interest income tax rule persistence	Beta	0.70	0.20
ρ_{td}	Dividend tax rule persistence	Beta	0.70	0.20
$\rho_{e\tau}$	Expensing rule persistence	Beta	0.70	0.20
<i>Fiscal rule debt-feedback coefficients</i>				
$\rho_{k,b}$	Capital tax / debt	Gamma	0.50	0.25
$\rho_{w,b}$	Labor tax / debt	Gamma	0.50	0.25
$\rho_{g,b}$	Government spending / debt	Gamma	0.50	0.25
$\rho_{tr,b}$	Transfers / debt	Gamma	0.50	0.25
$\rho_{c,b}$	Consumption tax / debt	Gamma	0.50	0.25
$\rho_{ti,b}$	Interest income tax / debt	Gamma	0.50	0.25
$\rho_{td,b}$	Dividend tax / debt	Gamma	0.50	0.25
$\rho_{ic,b}$	ITC / debt	Normal	0.00	0.25
$\rho_{e\tau,b}$	Expensing / debt	Normal	0.00	0.25
<i>Fiscal rule output-feedback coefficients</i>				
$\rho_{k,y}$	Capital tax / output	Gamma	0.15	0.10
$\rho_{w,y}$	Labor tax / output	Gamma	0.15	0.10
$\rho_{g,y}$	Government spending / output	Normal	-0.05	0.05
$\rho_{tr,y}$	Transfers / output	Normal	-0.10	0.05
$\rho_{c,y}$	Consumption tax / output	Normal	0.00	0.10
$\rho_{ic,y}$	ITC / output	Normal	0.00	0.10
$\rho_{ti,y}$	Interest income tax / output	Normal	0.00	0.10
$\rho_{td,y}$	Dividend tax / output	Normal	0.00	0.10
$\rho_{e\tau,y}$	Expensing / output	Normal	0.00	0.10
<i>Cross-shock correlations</i>				
ϕ	ε_t^k entering labor tax rule	Normal	0.25	0.10
ϕ_{wd}	ε_t^w entering dividend tax rule	Normal	0.25	0.10
<i>Shock standard deviations (all inv-Gamma, mean 0.05, std 0.10)</i>				
$\sigma_{\varepsilon^k}, \sigma_{\varepsilon^w}, \sigma_{\varepsilon^g}, \sigma_{\varepsilon^{tr}}, \sigma_{\varepsilon^z}, \sigma_{\varepsilon^d}, \sigma_{\varepsilon^\mu}, \sigma_{\varepsilon^m}$	Zubairy shocks	Inv-Gamma	0.05	0.10
$\sigma_{\varepsilon^c}, \sigma_{\varepsilon^{ic}}, \sigma_{\varepsilon^{ti}}, \sigma_{\varepsilon^{td}}, \sigma_{\varepsilon^{e\tau}}$	Extension shocks	Inv-Gamma	0.05	0.10

Note: Parameters below the horizontal rule are new to this paper relative to Zubairy (2010). Gamma priors for debt-feedback coefficients of tax rules enforce positive values (taxes rise with debt). Normal priors for debt-feedback coefficients of the ITC and expensing rules are centered at zero because these instruments may either rise or fall in response to debt. Gamma priors for output-feedback coefficients enforce the sign implied by automatic stabilizers for tax rates; Normal priors for spending rules allow the sign to be determined by the data.